
The Invisible Ledger

Quantifying the Invisible Wedge in Indonesia's Platform Economy

隱形帳簿：量化印尼平台經濟中的隱形缺口

CHRISTOPHER ONGKO 王新福

ADVISOR: PROF. DE-RONG KONG 孔德蓉

MASTER OF SCIENCE IN FINANCE, YUAN ZE UNIVERSITY · PROPOSAL ORAL EXAMINATION · 1 OCTOBER 2026

Digital commerce has changed how an entire country buys and sells, yet some of the numbers used to measure it make it look far smaller than others do.

Sellers in any village reach the whole country; shops run from phones; payments move by QR scan.

Reports on the same activity describe it as booming, as modest, or as barely growing.

Each report is correct on its own terms. Side by side, they don't tell the same story.

Case Study: Tanah Abang Market, Jakarta, 2023

“Everyone could see commerce moving online. No single number could say how much.”

2023

SEPTEMBER – OCTOBER 2023

DECEMBER 2023 – JANUARY 2024

THE SCENE

Tanah Abang in Jakarta is Southeast Asia's largest wholesale market. Its traders said their customers had gone elsewhere: to **sellers live-streaming on their phones**.

THE RESPONSE

The government restricted selling through social media. Within days, **TikTok Shop closed** in Indonesia.

THE RETURN

TikTok came back by **buying control of Tokopedia**, one of Indonesia's largest marketplaces, and folding its shop into it.

THE QUESTION

How much commerce had actually moved online? It depends on which number you read.

The platforms' accounts, their transaction figures and the national statistics **each saw a different part of the same shift**.

Sources: Trade Minister Regulation 31/2023; ABC News (5 Oct 2023); Al Jazeera (23 Oct 2023); CNN (11 Dec 2023); Nikkei Asia. Deal completed 31 January 2024.

One Economy, Several Numbers

“The numbers are built to differ in size. The question is whether they tell the same growth story.”

Different numbers, different parts

TRANSACTION VALUE (GMV / GTV):

All sales passing through a platform, within its stated reporting scope.

PLATFORM REVENUE:

Revenue recognized by the platform under its reporting model. May include fees, commissions and other monetization, depending on the business model and reporting treatment. **Investors value platforms largely on this.**

NATIONAL STATISTICS (BPS):

What surveyed businesses report selling online, by channel, including **social media and messaging**.

A large gap in size is **normal**: a platform may recognize only a fraction of transaction value as revenue. **None of these numbers is wrong.**

Where they can part: growth

Platform	Year	Transaction value	Revenue
Etsy	2021-22	-1.3%	+10.2%
eBay	2022-23	-0.9%	+3.2%
Zalando	2021-22	+3.2%	-0.1%

ACROSS 8 LISTED PLATFORMS ABROAD:

→ Revenue grew faster in **22** of 29 comparisons

→ The two moved in opposite directions in **4**

Growth stories can **split** even at platforms everyone knows. Abroad, the drift is **usually mild**.

Benchmark: eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify, Zalando · issuer filings, 2017–2025 · comparisons unaffected by acquisitions or scope changes.

Why Indonesia, and Why Tokopedia

“Everywhere the numbers drift. In Indonesia they split, and one company shows it clearly.”

Why Indonesia

WHY IT MATTERS:

Indonesia is the **largest digital market in Southeast Asia**, and one where commerce moved online quickly, through marketplaces, social media and messaging alike.

WHY IT IS TESTABLE:

Several local issuers disclose both revenue and transaction value over time, so the relationship can be tested **within the same firm**. BPS also reports national online sales by channel and the number of selling businesses, so platform evidence can be set against **national evidence**.

Meet Tokopedia

WHO:

One of Indonesia’s largest marketplaces, roughly its eBay, and part of GoTo alongside Gojek.

It is also the company **TikTok bought control of** in the Tanah Abang story.

THE PUZZLE, 2022 → 2023:

+53.2%

revenue

−8.9%

transaction value

Both figures come from **GoTo’s own disclosures**. Read through its accounts, Tokopedia was **expanding quickly**; read through its transactions, it was **shrinking**.

A Bigger Slice of a Smaller Pie

“Most of the revenue increase reconciles to lower incentives, not more commerce, and different users read that year differently.”

WHAT CHANGED (2022 → 2023):

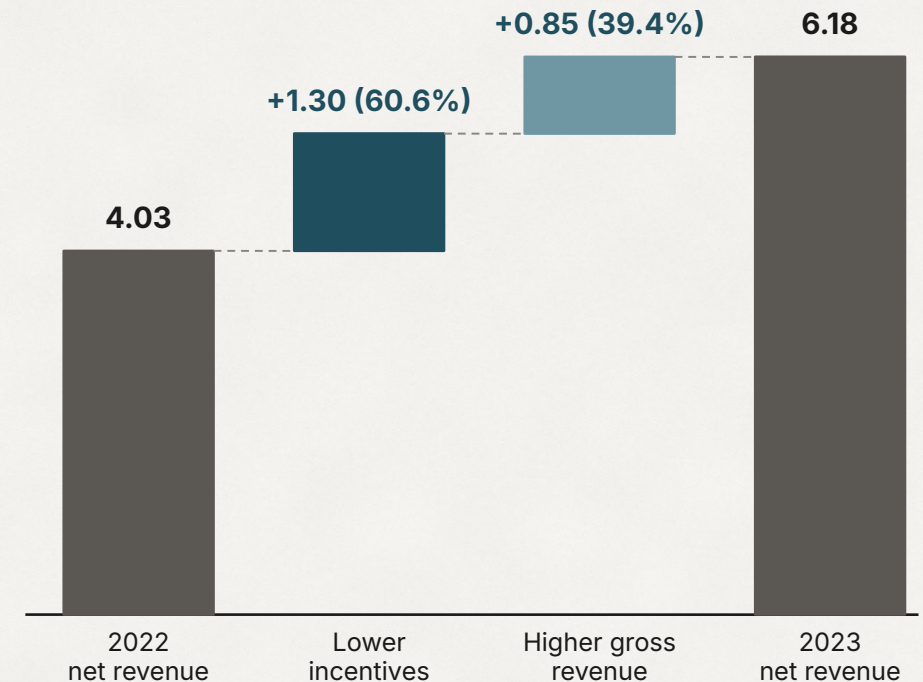
- Transaction value **fell**
- Net revenue **rose**
- Customer incentives deducted from gross revenue **fell** (discounts, vouchers, promotions)

THE RECONCILIATION:

- **60.6%** of the net-revenue increase reconciles to **lower incentives**
- 39.4% reconciles to higher gross revenue
- *Arithmetic reconciliation, not a causal claim*

WHY IT MATTERS: WHO READS WHICH NUMBER:

- **Investors** largely read revenue
- **Statisticians** read seller surveys
- **Tax administration** increasingly draws on marketplace records
- Same activity, same year: the conclusions can point in **opposite directions**



TOKOPEDIA NET REVENUE, Rp TRILLION

Gross revenue Rp8.14T → Rp8.99T · incentives Rp4.11T → Rp2.81T · net revenue
Rp4.03T → Rp6.18T (GoTo, 2024 annual report)

What the Literature Can and Cannot Answer

“Existing work explains why the measures differ; this study directly tests when substituting one for another changes the growth conclusion.”

What we know

PLATFORM ECONOMICS (Rochet & Tirole, 2003; Hagiu & Wright, 2015)

Platforms price through fees and subsidies across both sides of the market, so revenue **need not** track the commerce they coordinate.

ACCOUNTING (IFRS 15; Bowen, Davis & Rajgopal, 2002)

Principal/agent rules and incentives decide **how much of each sale becomes revenue**, and both can change.

KPI RESEARCH (Givoly et al., 2019)

A reported measure loses information when its **basis of computation changes** over time.

MEASUREMENT (Henderson, Storeygard & Weil, 2012; Ahmad & Schreyer, 2016)

An indicator must be **tested** against what it stands in for. National accounts separate a platform’s service from the commerce it carries.

DIGITAL RECORDS (Berg et al., 2020; OECD, 2020)

Platform records can **show activity that revenue excludes**, which is why tax rules increasingly draw on them.

What’s missing

UNANSWERED:

- *When* does revenue stop representing platform commerce?
- Does platform data represent *national* growth?
- How large do these differences become in practice, and do they persist?

WHY IT’S HARD:

It needs **both numbers from the same company, over time**, plus a **benchmark of normal drift**. Indonesia adds local issuers reporting both, *and* national surveys that split online sales by channel.

THE STUDY’S PLACE:

Directly tests the stability of the revenue–commerce relationship and links it to national measurement.

Research Question and Hypotheses

“Does revenue represent the platform, and does the platform represent the country?”

RESEARCH QUESTION:

How much economic activity remains invisible when digital platforms are measured through their reported revenue, and when does that choice change the conclusion about growth?

H1 — REVENUE → PLATFORM COMMERCE:

Transaction value and revenue recognized by the same platform **grow proportionally over time**.

*Why expect a test here: revenue tracks transactions **only while the share kept stays constant**, and fees, incentives and accounting treatment all move it.*

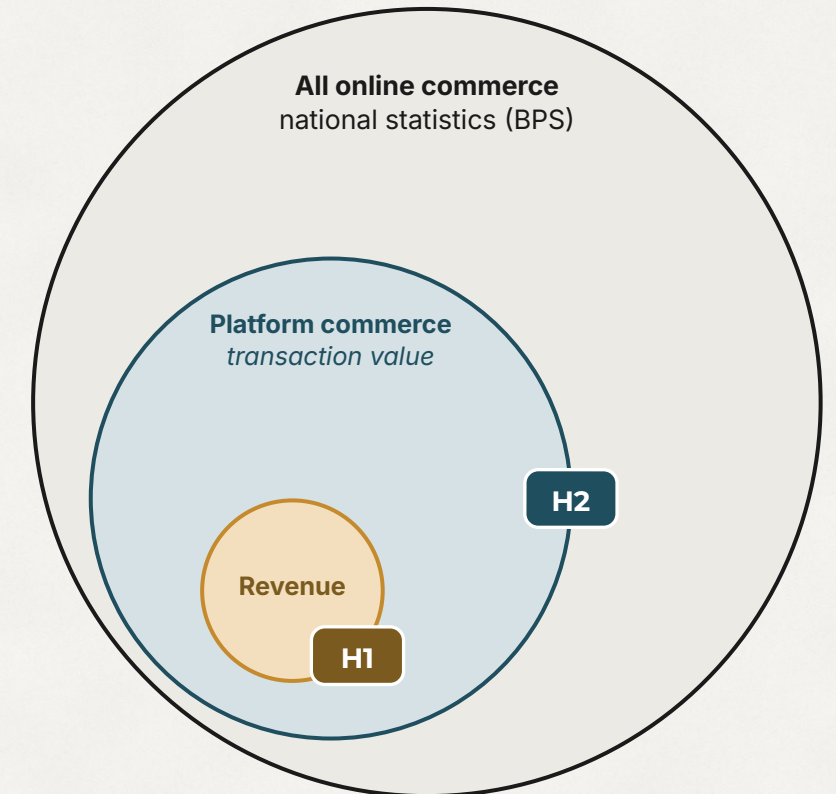
H2 — PLATFORM COMMERCE → NATIONAL GROWTH:

National e-commerce growth comes mainly from **more participating businesses**, not higher sales per business.

*Why it matters: growth can come from existing sellers selling more or from more sellers (Hummels & Klenow, 2005). Growth through more sellers can occur in **channels platform records don't cover**.*

SUPPORTING ANALYSES:

- Where national growth occurs, by sales channel
- Whether marketplace sellers keep better financial records



H1 (inner boundary): does revenue track platform commerce?
H2 (outer boundary): does platform commerce track national growth?
Illustrative only; circle sizes not to scale.

From Question to Measurement

“The size of the gap isn’t the test. Whether it stays stable is.”

THE LOGIC:

Revenue = **share kept** × **transaction value**. If the share **holds steady**, revenue and transactions grow together; if it moves, revenue tells a different growth story.

WHY D, NOT E:

E describes a level at one point in time, and levels differ with the business model. D asks **whether that level moves**, and that is what decides whether revenue growth still tracks commerce growth.

WHAT THE DATA MUST SHOW:

- **H1**: does revenue growth remain close to transaction-value growth within platforms?
- **Benchmark**: is the observed $|D|$ unusually large relative to comparable listed platforms?
- **H2**: does national growth come mainly from more participating businesses?

$$W = V - R$$

Invisible wedge: the level of the gap, in currency units

$$E = W / R = 1/m - 1$$

Ecosystem Ratio: the gap relative to revenue ($m = R/V$, the share kept)

$$D = g(R) - g(V)$$

GROWTH DIVERGENCE: THE CENTRAL TEST

$D \approx 0$ → revenue approximately preserves the transaction-growth signal

$|D|$ large → the share kept changed enough to alter the growth story

The Dataset: Sufficient, Not Unlimited

“Few platforms disclose both numbers, so the method is built around that constraint.”

What exists

INDONESIAN PLATFORMS:

Many platforms, few listed, **fewer reporting both numbers themselves**. Three do: **Tokopedia, Blibli and Bukalapak**, giving 11 platform-years and **8 year-to-year comparisons** (2020–2025).

HOW EACH SOURCE IS USED:

- **Main sample**: issuer-reported pairs → H1
- **Benchmark**: 8 listed platforms abroad, 29 clean comparisons → what “normal” looks like
- **BPS national surveys, 2022–2024** → H2
- **Grab, Shopee**: constructed Indonesian figures → 2023 scale only
- **Bank Indonesia payments** → context only

What this means for method

MATCHING RULE:

A pair enters **only when** transaction value and revenue cover the same period and, as closely as possible, the same business activity. **Mismatched pairs are excluded**; issuer labels (GTV, GMV, TPV) are kept and checked.

DESIGN CHOICES:

- Compare each platform **with itself** over time
- Benchmark Indonesia against **normal drift abroad**
- Check that no single firm or case drives the result

WHAT’S KEPT APART:

- Issuer-reported vs constructed figures
- Platform vs national data: different definitions, **never** pooled

Lazada and TikTok Shop are excluded: no matched Indonesian revenue series exists. Tokopedia’s series ends in 2023, when TikTok took control and folded its shop into it.

Scale: Large, but Not the Finding

“In 2023, transaction value across three platforms was about 13.7 times their revenue ($V/R \approx 13.7$; $E = W/R \approx 12.7$).”

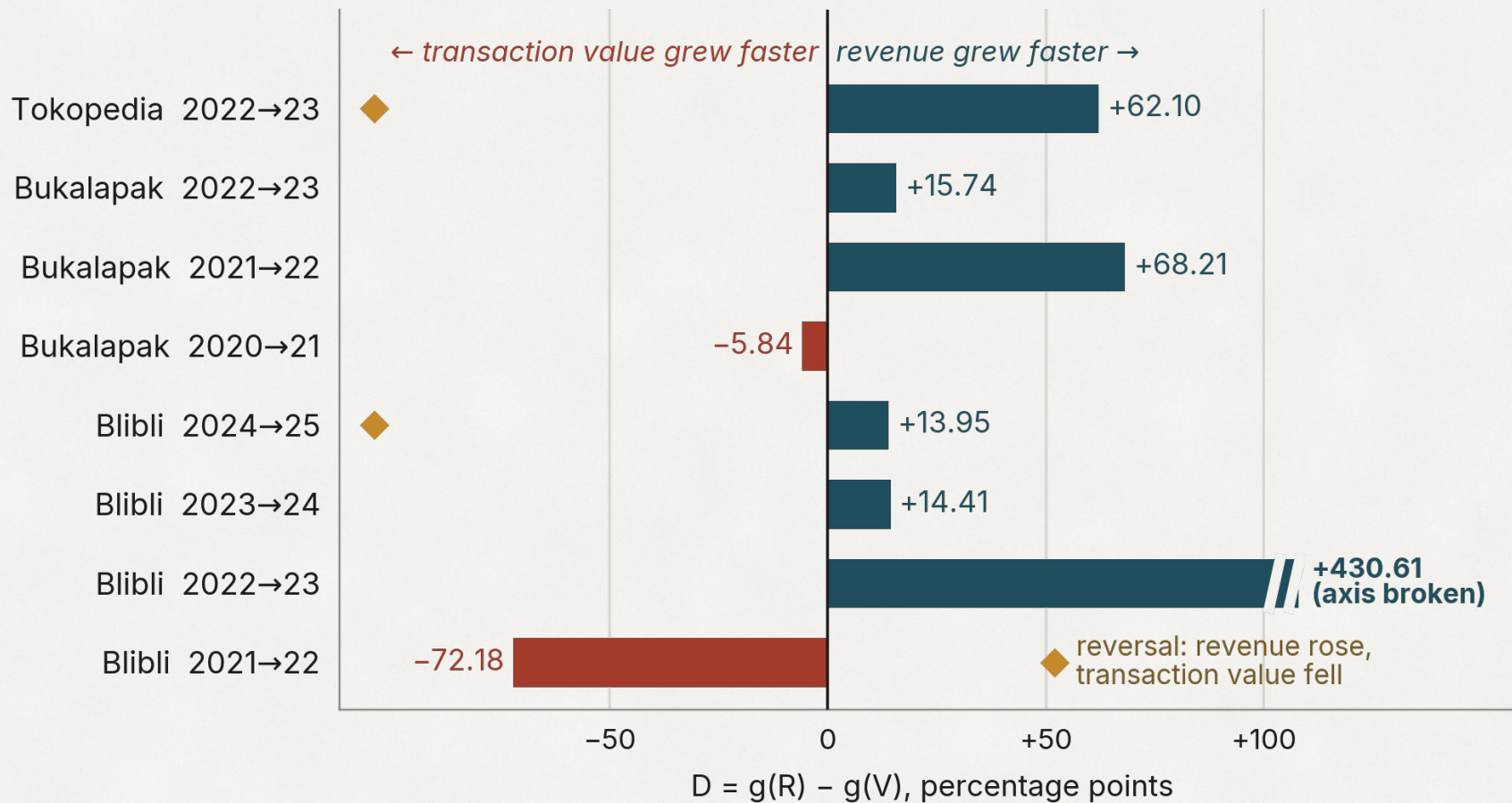
2023, US\$ billions	Transaction value (V)	Revenue (R)	Wedge (W)	Ratio (E)	Evidence
Tokopedia	16.33	0.41	15.93	39.3	Issuer-reported
Shopee	21.52	2.15	19.37	9.0	<i>Constructed</i>
Grab	5.38	0.61	4.78	7.9	<i>Constructed</i>
Total	43.23	3.16	40.07	12.7	

The wedge matters for anyone inferring the *level* of commerce from revenue. But a large, *stable* wedge would **not by itself** make revenue a misleading guide to *growth*.
W and E tell us whether the levels differ. D tells us whether that difference **changes the growth story**, and that is what the next slides test.

Shopee's assumed rate of 9–11% gives US\$39.9–40.3B in total; dropping any one platform still leaves ≥ US\$20.7B. Grab's and Shopee's ratios reflect company-wide figures by construction.

H1: The Pattern

“Revenue grew faster than transaction value in 6 of 8 comparisons; in 2 of them, revenue rose while transaction value fell.”



READING IT:

Each bar is one platform-year comparison: revenue growth minus transaction-value growth.

THE PATTERN:

Median gap **38.92 percentage points**.

Two reversals, where revenue rose while transaction value fell: Tokopedia 2022-23 and Blibli 2024-25.

SO:

Tokopedia **wasn't a one-off**. But growth rates never match exactly: **is this unusual?**

H1: Preliminary Formal Evidence

“Indonesia’s divergence is more than five times the benchmark, and it survives each restriction tested.”

	Indonesia	Abroad
Median $ D $	38.92 pp	7.10 pp
Growth correlation	0.10	0.98 (excl. Sea)
Comparisons	8	29

WHAT SETS INDONESIA APART:

Not the direction: abroad, revenue also tends to grow a little faster than transactions. The difference is **the size of the gap**, and **how weakly the two growth rates move together**.

THE TEST:

→ One-sided Mann–Whitney on $|D|$: **p = 0.002**

ROBUSTNESS:

→ Drop the largest case → p = 0.006

→ Exclude Sea from the benchmark → p < 0.001

→ One median per firm → p = 0.024

INTERPRETATION:

→ No single statistic carries it; the evidence is cumulative

→ **Evidence inconsistent with H1**

Benchmark: 29 clean comparisons from 8 listed platforms outside Indonesia (issuer filings, 2017–2025). Sea excluded where noted because its Shopee business also operates in Indonesia.

H2: Preliminary National Evidence

“National growth came mainly from more participating businesses, not higher sales per business.”

BPS e-commerce survey	2022-23	2023-24
E-commerce value	+40.60%	+17.08%
Businesses	+27.40%	+15.31%
Sales per business	+10.36%	+1.54%
Share from more businesses	≈71%	≈90%

HOW IT IS SPLIT:

National e-commerce can grow because **more businesses sell online**, or because **each business sells more**. BPS reports both, so growth can be divided between the two.

FINDING:

Growth came mainly from an increase in participating businesses, **in both years**. BPS’s alternative 2023 count changes the **size, not the conclusion**.

INTERPRETATION:

Evidence consistent with H2 (an arithmetic decomposition, not entry dynamics).

SUPPORTING CHANNEL RESULT, 2023-24

Marketplace category +1.45% · other channels +20.57% → **98.46% of measured growth outside BPS’s marketplace category**

*Why it matters: growth through more sellers can take place in channels platform accounts do not cover, the kind of channels the Tanah Abang case involved, so platform figures can miss it even when **every platform reports accurately**.*

2023: 95.33% of e-commerce businesses used instant messaging · complete financial statements: 28.63% of marketplace users vs 12.25% of non-users (association, not cause).

Platform and National Measures Are Not Yet Reconcilable

“Their levels differ sharply, and they classify economic activity differently; reconciling those definitions is part of the thesis.”

Levels, 2023

BPS MARKETPLACE CATEGORY:

About US\$13.2B (Rp200.68T), **18.2%** of national e-commerce value.

PLATFORMS' OWN FIGURES:

Tokopedia's transaction value plus Shopee's estimate: about US\$37.9B. **Either one alone is larger** than the BPS marketplace category.

WHY THEY ARE NOT YET LIKE-FOR-LIKE:

- BPS assigns surveyed online sales to statistical channels
- Platform disclosures aggregate transaction activity under each platform's **own scope and definitions**
- Coverage, attribution and reporting scope therefore require reconciliation

WHAT THE THESIS DOES FIRST:

Reconcile the definitions before attributing any part of the difference to coverage.

Growth, 2023 → 2024

E-COMMERCE (BPS): +17.08%

PAYMENTS (BANK INDONESIA):

- E-money shopping +30.47%
- Mobile banking +82.84%
- QRIS +186.98%

Each measurement system tells **its own growth story**. Payments also reflect adoption and non-commerce use, so they are context, **not sales estimates**.

Robustness and Boundaries

“Strong conclusions, drawn inside clear limits.”

Questions we expect

Q: Is it one outlier?

A: No. Without the largest case (Blibli 2022–23, +430.61 pp), the test still gives **p = 0.006**.

Q: Aren't a firm's years dependent?

A: Yes, so the test is repeated with **one median per firm** ($p = 0.024$).

Q: Do the constructed figures drive it?

A: No. Grab and Shopee are **excluded from H1**.

Q: Aren't these just different business models?

A: The benchmark deliberately spans very different platform models. Despite that, its median $|D|$ is **7.1 pp against 38.9 pp** in Indonesia; using growth rather than levels also limits level-comparability concerns.

Q: Is it causal?

A: No. This is a **measurement and comparability study**.

Boundaries

DOES NOT IMPLY:

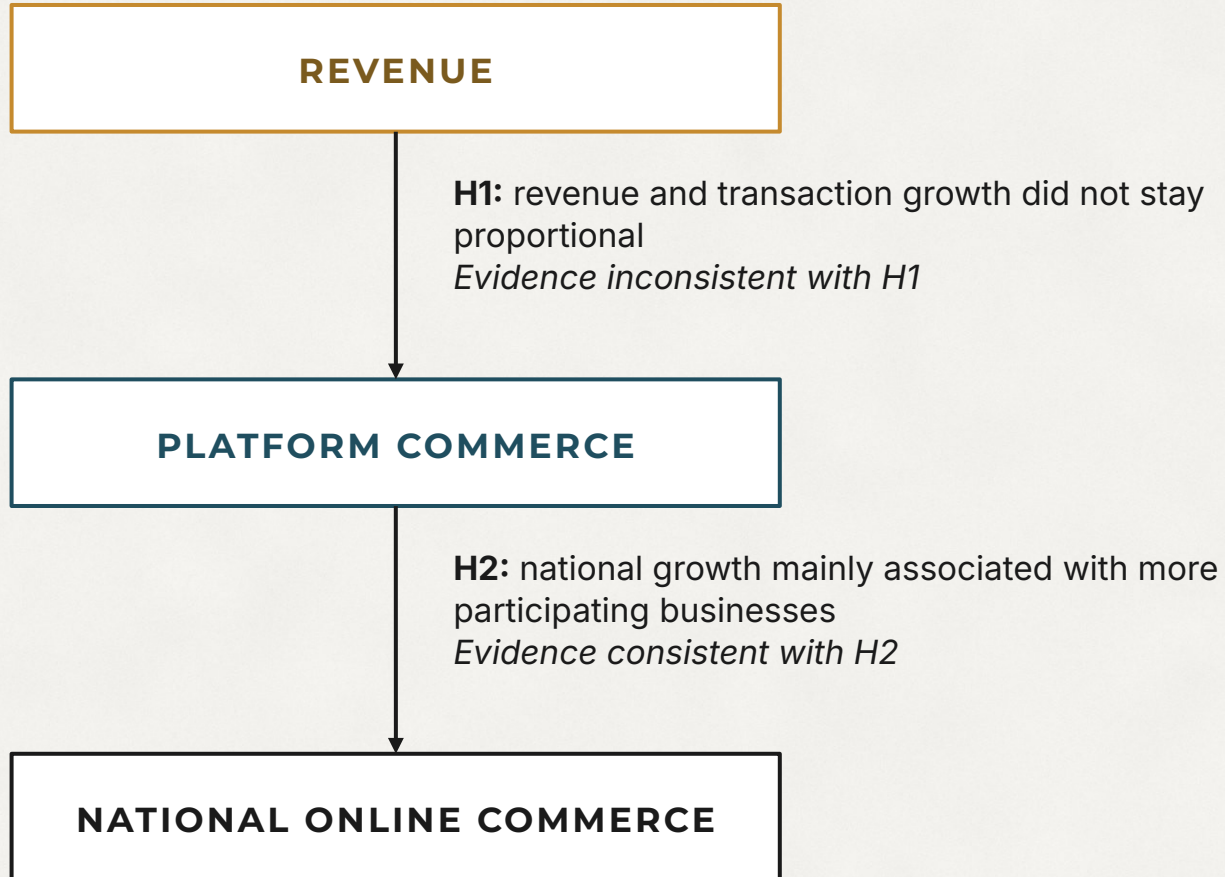
- Hidden GDP
- Unpaid tax
- Investor mispricing
- Error in any one indicator

LIMITS:

- Three firms disclose both numbers
- A purposive benchmark, not a random sample
- Two BPS transitions (one for the channel split)
- Levels across systems not yet reconciled

The Complete Chain

“Each link from the reported number to the economy can lose information.”



SUPPORTING CHANNEL RESULT

98.46% of measured 2023–24 growth occurred outside BPS's marketplace category

EVIDENCE SO FAR

- ✓ Within-platform divergence
- ✓ Benchmark test
- ✓ National decomposition
- ✓ Channel split
- Level reconciliation (thesis)

Contribution

“What this study adds so far, and a check to run before one number stands in for another.”

What this study adds so far

Open question	What this study adds so far
When does revenue stop representing platform commerce?	A tracked measure (D), benchmarked abroad: in Indonesia, revenue and transaction growth diverge far more than abroad
How large are the differences, and do they persist?	Large in level and moving over time; at least one major divergence reconciles to disclosed components (Tokopedia: incentives); the thesis will diagnose the others
Does platform data represent national growth?	National growth came mainly from more selling businesses, largely outside BPS's marketplace category

FOR FINANCE:

Revenue growth can come from **more commerce** or from **keeping a bigger share** of each sale. In revenue they look identical; **they need not carry the same information**.

The framework: a check before substituting

- 1 MATCH**
Pair transaction value (V) and revenue (R) for the same period and the same business activity.
- 2 SIZE THE GAP**
 $W = V - R$ and $E = W / R$: how much sits outside revenue. Levels differ by business model.
- 3 TEST STABILITY**
 $D = g(R) - g(V)$. Near zero: revenue approximately preserves the growth signal. Large $|D|$: growth materially diverges, consistent with a change in monetization.
- 4 BENCHMARK AND DIAGNOSE**
Compare $|D|$ with comparable platforms; diagnose large divergences using disclosed incentives, fees and scope.

WHAT THE THESIS WILL COMPLETE: thicken H1 with quarterly pairs and a longer benchmark · diagnose each large divergence · reconcile platform and BPS definitions · *market test: does the stock market tell the two kinds of revenue growth apart?*